Arindam Khanda

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Arindam Khanda • Arindam Khanda

I am a postdoctoral researcher at the CReWMaN Lab in the Department of Computer Science at Missouri University of Science and Technology (Missouri S&T), working under the mentorship of Prof. Sajal K. Das. My research focuses on ***high-performance computing*** (***HPC***) ***for graph analytics, scalable AI***, and their applications in ***cyber-physical systems*** (***CPS***). I was recognized with the ***2025 NSF CPS Rising Stars award*** for contributions to cyber-physical systems using dynamic graph analysis. I also serve as Student Program Co-Chair for ***ACM International Symposium on High-Performance Parallel and Distributed Computing*** (***HPDC***) ***2026***.

Research Interests

High-Performance Data Analytics, Parallel and Distributed Computing, Graph Analytics, Scalable Artificial Intelligence, Heterogeneous Computing, Cyber-Physical and Data-intensive Systems

Education

- **Missouri University of Science and Technology** — **USA**
  Ph.D. in Computer Science, CGPA: 3.97/4 — *Aug 2019 – May 2024*
  Thesis Title: Parallel Algorithms for Large Scale Dynamic Graph Analysis
  Advisor: Prof. Sajal K. Das.
- **Birla Institute of Technology & Science (BITS), Pilani** — **India**
  M.Tech. (WILP) in Software Systems (Specialization: Data Analytics), CGPA: 9.45/10 — *Jan 2017 – Dec 2018*
- **Maulana Abul Kalam Azad University of Technology (formerly known as WBUT)** — **India**
  B.Tech. in Electronics & Communication Engineering, CGPA: 9.04/10 — *2011 – 2015*

Professional Appointments

- **Missouri University of Science and Technology, USA** — **MO, USA**
  Postdoctoral Fellow, CReWMaN Lab, Department of Computer Science — *Jun 2024–Present*
  Mentor: Prof. Sajal K. Das
- **Missouri University of Science and Technology, USA** — **MO, USA**
  Graduate Research Assistant — *Fall 2019 - Spring 2024*
- **Argonne National Lab** — **IL, USA**
  W. J. Cody Summer Intern — *May 2023–Aug 2023*
  Mentor: Michael Kruse
- **Argonne National Lab** — **IL, USA**
  W. J. Cody Summer Intern — *May 2022–Aug 2022*
  Mentor: Getnet Betrie
- **Infosys Limited** — **Bhubaneswar, Odisha, India**
  Technology Analyst — *Dec 2018–Jul 2019*
- **Ericsson** — **Kolkata, West Bengal, India**
  Senior Solution Integrator — *Nov 2015–Nov 2018*

Publications

Journals

[1] S. M. Shovan*, **A. Khanda***, and S. K. Das, "Parallel Multi Objective Shortest Path Update Algorithm in Large Dynamic Networks," *IEEE Transactions on Parallel and Distributed Systems*, vol. 36, no. 5, pp. 932–944, 2025. https://doi.org/10.1109/TPDS.2025.3536357 (***Equal contribution.**).

[2] **A. Khanda**, S. Srinivasan, S. Bhowmick, B. Norris, and S. K. Das, "A Parallel Algorithm Template for Updating Single-Source Shortest Paths in Large-Scale Dynamic Networks," *IEEE Transactions on Parallel and Distributed Systems*, vol. 33, no. 4, pp. 929–940, 2022. https://doi.org/10.1109/TPDS.2021.3084096.

[3] **A. Khanda**, A. Satpathy, and S. K. Das, "Smart-charge: Stable matching algorithm for electric vehicle charging in subscription-based models," *Pervasive and Mobile Computing*, vol. 118, p. 102197, 2026.

Refereed Conference Papers

[1] S. M. Shovan, **A. Khanda**, S. M. Ferdous, S. K. Das, and M. Halappanavar, "DynLP: Parallel Dynamic Batch Update for Label Propagation in Semi-Supervised Learning," in *2026 ACM International Conference on Supercomputing* (*ICS*), 2026. (Accepted For Publication).

[2] S. M. Shovan*, **A. Khanda***, S. Bhowmick, and S. K. Das, "ESCHER: Efficient and Scalable Hypergraph Evolution Representation with Application to Triad Counting," in *2026 IEEE International Parallel and Distributed Processing Symposium* (*IPDPS*), 2026. (***Equal contribution.**)(Accepted For Publication) https://arxiv.org/abs/2512.21009.

[3] A. Satpathy*, **A. Khanda***, C. Swain, and S. K. Das, "ReACT-TTC: Capacity-Aware Top Trading Cycles for Post-Choice Reassignment in Shared CPS," in *Proceedings of the ACM/IEEE 17th International Conference on Cyber-Physical Systems* (*ICCPS*), 2026. (***Equal contribution.**)(Accepted For Publication) https://arxiv.org/abs/2602.00859.

[4] S. Tammali, **A. Khanda**, A. Satpathy, S. Shovan, and S. K. Das, "RESCUE: Routing under Evolving Stochastic Congestion and Uncertain Spread in Wildfire Emergencies," in *Proceedings of the 27th International Conference on Distributed Computing and Networking* (*ICDCN*), pp. 168–172, 2026.

[5] **A. Khanda**, A. Satpathy, A. Jha, and S. K. Das, "CARGO: A Co-Optimization Framework for EV Charging and Routing in Goods Delivery Logistics," in *2025 IEEE 50th Conference on Local Computer Networks* (*LCN*), pp. 1–9, IEEE, 2025.

[6] D. Sengupta, A. Satpathy, **A. Khanda**, and S. K. Das, "CITRUS: Cost and Ischemia Time Reduction using Urban Air Mobility Solutions for Organ Transport," in *2025 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events* (*PerCom Workshops*), pp. 170–175, 2025. https://doi.org/10.1109/PerComWorkshops65533.2025.00060.

[7] **A. Khanda**, A. Satpathy, A. Vangala, and S. K. Das, "SMEVCA: Stable Matching-based EV Charging Assignment in Subscription-Based Models," in *Proceedings of the 26th International Conference on Distributed Computing and Networking* (*ICDCN*), pp. 46–55, 2025. https://doi.org/10.1145/3700838.3700851 (**Best Paper Nomination**).

[8] **A. Khanda**, S. Roy, P. Roy, and S. K. Das, "Structural Hole Spanners Detection in Directed Social Networks: A Feed Forward Loop Motif Approach," in *IEEE Global Communications Conference* (*GLOBECOM*), IEEE, 2024. https://doi.org/10.1109/GLOBECOM52923.2024.10901271.

[9] **A. Khanda**, S. M. Shovan, , and S. K. Das, "A Parallel Algorithm for Updating a Multi-objective Shortest Path in Large Dynamic Networks," in *Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis* (*SC-W 2023*), 2023. https://doi.org/10.1145/3624062.3625134.

[10] S. Srinivasan, **A. Khanda**, A. Pandey, S. Das, S. Bhowmick, and B. Norris, "A Distributed Algorithm for Identifying Strongly Connected Components on Incremental Graphs," in *2023 IEEE 35th International Symposium on Computer Architecture and High Performance Computing* (*SBAC-PAD*), pp. 109–118, IEEE, 2023. https://doi.org/10.1109/SBAC-PAD59825.2023.00020.

[11] **A. Khanda**, S. Bhowmick, X. Liang, and S. K. Das, "Parallel Vertex Color Update on Large Dynamic Networks," in *2022 IEEE 29th International Conference on High Performance Computing (HiPC)*, 2022. https://doi.org/10.1109/HiPC56025.2022.00027 **(Best Paper Nomination)**.

[12] **A. Khanda** and S. K. Das, "A Parallel Framework for Efficiently Updating Graph Properties in Large Dynamic Networks," in *Proceedings of the 24th International Conference on Distributed Computing and Networking*, ICDCN '23, (New York, NY, USA), p. 298–299, Association for Computing Machinery, 2023. https://doi.org/10.1145/3571306.3571359.

[13] **A. Khanda**, F. Corò, and S. K. Das, "Drone-Truck Cooperated Delivery Under Time Varying Dynamics," in *Proceedings of the 2022 Workshop on Advanced Tools, Programming Languages, and PLatforms for Implementing and Evaluating Algorithms for Distributed Systems*, ApPLIED '22, (New York, NY, USA), p. 24–29, Association for Computing Machinery, 2022. https://doi.org/10.1145/3524053.3542743.

[14] **A. Khanda**, F. Corò, F. B. Sorbelli, C. M. Pinotti, and S. K. Das, "Efficient Route Selection for Drone-based Delivery Under Time-varying Dynamics," in *2021 IEEE 18th International Conference on Mobile Ad Hoc and Smart Systems (MASS)*, pp. 437–445, 2021. https://doi.org/10.1109/MASS52906.2021.00061.

In Submission

[1] **A. Khanda**, "Application-aware decision making for modern electrified mobility systems," in *Secure and Responsible AI in 6G Technology-Enabled IoT and IoMT Systems*, CRC Press, Taylor and Francis Group, 2026. (Book Chapter)(Under review).

[2] **A. Khanda**, A. Satpathy, A. Jha, F. Betti Sorbelli, and S. K. Das, "CONVOY: Co-Optimizing Routing and Charging Decisions for EV Fleets in Urban Delivery Systems," *IEEE Transactions on Intelligent Transportation Systems*. (Under review).

[3] S. Raj, **A. Khanda**, R. Mittal, S. K. Das, and Y. Simmhan, "Mobility-Aware Cost-Efficient Co-Scheduling of UAV Routes and Analytics across Edge and Fog Computing," (Under double blind review).

[4] **A. Khanda**, S. Roy, P. Roy, and S. K. Das, "A Parallel Algorithm for Finding Robust Spanners in Large Social Networks," *IEEE Transactions on Parallel and Distributed Systems*. (Under Review) arXiv preprint arXiv:2508.01485.

[5] S. Srinivasan, **A. Khanda**, S. K. Das, B. Norris, and S. Bhowmick, "When the Graph Never Sleeps: Scalable Parallel PageRank for Evolving Networks," *Journal of Parallel and Distributed Computing*. (Under Review).

Research Experience

- **Missouri University of Science and Technology** **MO, USA**
  Postdoctoral Researcher *Jun 2024–Present*
  - Mentoring a Ph.D. student and co-developing GPU-efficient data structures and algorithms that enable scalable hypergraph analytics.
  - Developing a label-propagation-based approach for scalable semi-supervised learning on large, time-varying data.
  - Led the development of parallel methods for multi-objective shortest path search in *large dynamic networks*, leveraging heterogeneous computing architectures.
  - Co-developing routing and decision-making algorithms that integrate predictive models and optimization for *Urban Air Mobility* (UAM)-based organ transport, addressing vehicle availability, cold ischemia time, and flying costs.
  - Led the development of smart *electric vehicle (EV) charging* assignment solutions based on graph analytics, game theory, and mixed integer linear programming.
  - Co-developed an NSF proposal on *cyberinfrastructure for hypergraph-based analysis and innovation*, in collaboration with Prof. Sajal K. Das, and Dr. Sanjukta Bhowmick. (Submitted)

- **Argonne National Lab** **IL, USA**
  W.J. Cody Summer Intern *May 2023–Aug 2023*
  - Conducted HPC benchmarking research to evaluate efficiency, scalability, and suitability of HPC architectures for diverse workloads.
  - Enhanced the Rosetta benchmarking tool by adding SYCL support and enabling performance evaluation on SYCL-enabled heterogeneous computing architectures.
  - Developed and integrated a SYCL-optimized version of PolyBench into the benchmarking tool for improved assessment of HPC workloads.

- **Missouri University of Science and Technology** **MO, USA**
  Graduate Research Assistant *Fall 2019-Spring 2024*
  - Designed parallel dynamic graph algorithms, including shortest paths, vertex coloring, and strongly connected components, for large-scale dynamic network analytics.
  - Developed a generic update framework that reuses previously computed values and incremental network changes to efficiently update time-varying graph properties.
  - Leveraged NVIDIA CUDA, SYCL, OpenMP, and Open MPI to develop implementations optimized for diverse HPC architectures.
  - Contributed to NSF-funded projects: SANDY (Sparsification-Based Approach for Analyzing Network Dynamics) and CANDY (Cyberinfrastructure for Accelerating Innovation in Network Dynamics).

- **Argonne National Lab** **IL, USA**
  W.J. Cody Summer Intern *May 2022–Aug 2022*
  - Developed a deep learning framework for parameter estimation in large-scale brain networks using a Neuron-Synapse model represented by nonlinear ODEs.
  - Designed a two-model approach with a forward model to generate synthetic membrane potential data and an inverse model to predict Neuron-Synapse parameters from observed data.
  - Demonstrated framework efficiency using a Leaky Integrate-and-Fire (LIF) model with Spike-Timing-Dependent Plasticity (STDP) for learning image sequences in a spiking neural network.

Teaching Experience

Experienced in teaching algorithms, databases, cloud computing and big data, cyber-physical systems and graph theory at undergraduate and graduate levels.

- **Missouri University of Science and Technology** **Missouri, USA**
  Postdoctoral Fellow *Jun 2024–Present*
  - Guest lectured and supported instruction in:
    - Advances in Cyber-Physical Systems (Graduate)
    - Graph Theory (Graduate)
  - Designed and delivered talks on applications of dynamic graphs in CPS and parallel graph algorithms.
  - Mentored Ph.D. students on research methods, presentation skills, and publication preparation.

- **Missouri University of Science and Technology** **MO, USA**

  Graduate Teaching Assistant (*Database Optimization, CS 5300*) *Fall 2023*

  Graduate Teaching Assistant (*Algorithm, CS 2500*) *Spring 2023*

  Graduate Teaching Assistant (*Cloud Computing and Big Data Management, CS 6304*) *Fall 2022*

  Graduate Teaching Assistant (*Advances in Cyber-Physical Systems, CS 6001*) *Spring 2022*

  During my Ph.D., I gained extensive teaching experience across undergraduate and graduate courses.
  - Led recitation sessions for *Algorithms* (*CS 2500*), and fostered problem-solving skills among students.
  - In *Database Optimization* (*CS 5300*), I guided students through query optimization, prepared assignments, and designed final projects.
  - I provided hands-on instruction in big data technologies during *Cloud Computing and Big Data Management* (*CS 6304*). I set up cloud environments for all students and evaluated assignments.

- I supported students in research and project development in *Advances in Cyber-Physical Systems* (*CS 6001*), integrating my expertise in CPS applications.

External Funding & Proposal Activity

- **NSF CISE/OAC (CSSI Track) Proposal** **USA**
  Role: *Co-Principal Investigator,* ***Status: Funded*** *Apr 2026*
  Project: CHAI: cyberinfrastructure for hypergraph-based analysis and innovation.
  Projected Total Budget: $600,000
  – Prepared a collaborative project between Missouri S&T and the University of North Texas to enable scalable hypergraph analytics and AI on complex relational data.
  – Submitted to the NSF CSSI Elements program for a 3-year project.
  – Co-led data structure development for large dynamic hypergraphs.

- **NSF ENG/ECCS (EPCN Track) Proposal** **USA**
  Role: *Co-Principal Investigator,* ***Status:*** *Submitted* *Feb 2026*
  Project: SPICE: Subscription-based Platform for Intelligent Charging of Electric Vehicles.
  Projected Budget: $675,000
  – Prepared a proposal on equitable, resilient, and scalable EV charging infrastructure for smart cities.
  – Co-led system design, mobility modeling, and data-driven charging recommendation strategies; developing prototype evaluation plan using real EV data and city-scale mobility scenarios.

- **NSF National I-Corps Program** **USA**
  Principal Investigator and Technical Lead *Fall 2026 (planned)*
  Project: Time-efficient EV Trip Planner with Cost-efficient Charging (Sparks).
  Planned Award Target: $50,000
  – Completed NSF I-Corps Local + Regional Programs and led initial discovery efforts with a team, including a Ph.D. student, to identify pain points in current EV charging and EV-specific trip planning.
  – Leading commercialization expansion for a time-efficient EV trip planner, including large-scale customer discovery, market validation, and technology-readiness planning.
  – Plan to complete national I-Corps program to refine go-to-market strategy and prepare for SBIR/STTR.

Industrial Experience

- **Infosys Limited** **Bhubaneswar, Odisha, India**
  Technology Analyst *Dec 2018 — July 2019*
  - Contributed to a smart-grid billing application development as a Java developer.

- **Ericsson** **Kolkata, West Bengal, India**
  Senior Solution Integrator *Nov 2015 — Nov 2018*
  - Developed customer adaptation logic for Java-based Ericsson Product.
  - Led the development of two Robotic Process Automation (RPA)-based tools for web scraping, automated system monitoring, and reporting.
  - Contributed to troubleshooting and Root Cause Analysis (RCA) of Ericsson Multi Activation, a cross-platform provisioning solution by Ericsson.

Mentoring Experience

- S M Shovan | Missouri S&T — Ph.D. Candidate in Computer Science
- Sowjanya Tammali | Missouri S&T — Ph.D. Candidate in Computer Science
- Debjyoti Sengupta | Missouri S&T — Ph.D. Candidate in Computer Science
- Nabadip Deb | Ex-Ericsson — Customer Experience Leader, vidaXL

Poster Presentations and Talks

[1] A. Khanda, "Fantastic graph dynamics and where to find them: How to tame them and when to tame them," in *SC25 Birds of a Feather on HPC Graph Toolkits and the GraphBLAS Forum*, 2025. (BoF Presentation).

[2] A. Pandey, A. Khanda, S. M. Shovan, A. Y. Khan, B. Norris, S. K. Das, and S. Bhowmick, "Algorithms and Applications of Dynamic Network Analysis using CANDY," in *the International Conference on High Performance Computing, Network, Storage, and Analysis (SC'25)*, 2025. (**Best Poster Nomination**).

[3] A. Khanda, "Efficient Techniques in Analyzing Large Dynamic Networks ," in *SIAM Conference on Computational Science and Engineering (CSE25)*, 2025.

[4] S. Bhowmick, A. Khanda, and S. K. Das, "Applications of Dynamic Graph Analysis using CANDY," in *NSF CSSI PI Meeting*, 2025.

[5] A. Khanda, "Electrifying Urban Transportation: Exploring Routing and Charging Challenges ," in *NSF CPS PI Meeting (NSF CPS Rising Stars)*, 2025.

[6] A. Khanda, S. M. Shovan, A. Pandey, F. Hosseini, S. K. Das, B. Norris, and S. Bhowmick, "Efficient Approaches to Analyzing Large Dynamic Networks," in *the International Conference on High Performance Computing, Network, Storage, and Analysis (SC'24)*, 2024.

[7] A. Pandey, A. Khanda, S. Srinivasan, S. Srinivasan, S. M. Shovan, F. Hosseini, S. K. Das, B. Norris, and S. Bhowmick, "Scalable Algorithms for Analyzing Large Dynamic Networks Using CANDY," in *the International Conference on High Performance Computing, Network, Storage, and Analysis (SC'23)*, 2023.

[8] A. Pandey, A. Khanda, S. Srinivasan, S. Bhowmick, S. K. Das, and B. Norris, "CANDY: An Efficient Framework for Updating Properties on Large Dynamic Networks," in *the International Conference on High Performance Computing, Network, Storage, and Analysis (SC'22)*, 2022.

[9] A. Khanda and S. K. Das, "Efficient Parallel Algorithm for Shortest Path Updates in Dynamic Networks at Scale," in *ACM-SRC in the International Conference on High Performance Computing, Network, Storage, and Analysis (SC'21)*, 2021. (ACM Student Travel Grant).

[10] S. Srinivasan, A. Pandey, A. Khanda, S. Srinivasan, and S. K. Das, "Parallel Framework for Updating Large-Scale Dynamic Networks," in *the International Conference on High Performance Computing, Network, Storage, and Analysis (SC'21)*, 2021.

Professional Services and Leadership

Workshop co-chair, organizing committee member, and TPC member for conferences including HPDC, HiPC, ICDCN, and workshops co-located with MASS, DCOSS IoT, and ICPP.

- ICDCN (2027) — Social Media Chair
- ACM HPDC (2026) — Student Program Co-Chair
- IEEE MASS (2026) — TPC Member
- LCN (2026) — TPC Member
- GLOBECOM 2026 MLCN Symposium — TPC Member
- Wi-DroIT in conjunction with DCOSS-IoT (2026) — TPC Member
- CREATIVE Workshop in conjunction with IFIP Networking (2026) — TPC Member
- ASCENT in conjunction with ICDCN (2026) — Organizing Co-Chair
- COMSNETS (2026) — TPC Member (Poster Session)
- HiPC (2025) — TPC Member (Student Research Symposium)
- Supercomputing India (SCI 2025) — TPC Member
- UAV-IoT in conjunction with IEEE MASS (2025) — Publicity Chair and TPC Member
- Wi-DroIT in conjunction with DCOSS-IoT (2025) — Workshop Co-Chair
- GRAND in conjunction with ICPP (2025) — Organizing Committee Member
- SANDY in conjunction with ICPP (2023 and 2024) — Organizing Committee Member

- The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'21, SC'22, SC'23) — Student Volunteer
- IEEE TPDS, TSC, TSUSC, T-ITS — Reviewer
- JPDC, FGCS, Ad Hoc Networks, JCST — Reviewer
- BigData Conference, IEEE ANTS (2024), HiPC (2025), SCI (2025), DCOSS-IoT (2026) — Reviewer

Technical Skills

- **Programming Languages:** Python, C++, Java, SQL
- **High-Performance Computing:** CUDA C++, OpenMP, OpenMPI, SYCL
- **Machine Learning:** TensorFlow, Keras, scikit-learn
- **Others:** MongoDB, Apache Hadoop, Spark, Winautomation, Git, LaTeX, CMake, Gurobi

Honors and Awards

2025 SC'25 Early Career Program Travel Grant
2025 SC'25 Best Research Poster Nomination
2025 NSF CPS Rising Stars award
2025 ICDCN'25 best paper nomination
2023 Missouri S&T Dean's Ph.D. Scholar Award
2023 Council of Graduate Studies (CGS), Missouri S&T best research poster award
2023 ICDCN'23 best Ph.D. proposal award under the doctoral symposium
2023 ICDCN'23 NSF Travel Grant
2023 AKC100 symposium'23 best poster award
2022 HiPC'22 best paper nomination
2022 Missouri S&T Graduate Student Travel Grant
2021 SC'21 ACM SRC Travel Grant
2017 Ericsson Ace Award
2016 Ericsson Ace Award

References

- **Prof. Sajal K. Das**
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  Department of Computer Science
  Missouri University of Science and Technology
  Rolla, MO 65409, USA
  Email: sdas@mst.edu
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- **Dr. Sanjukta Bhowmick**
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  Department of Computer Science and Engineering
  University of North Texas
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- **Dr. Francesco Betti Sorbelli**
  Associate Professor
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